

8

7

$$W_{balance} - T = m_{balance} a$$

$$(800)(9.81) - T = (800)a \text{ --- (1)}$$

$$T - W_{car+man} = m_{car+man} a$$

$$T - (600)(9.81) = 600a \text{ --- (2)}$$

(1) + (2):

$$(200)(9.81) = 1400a$$

$$a = 1.4 \text{ m s}^{-2}$$

8

$$F = N\rho Av^2 = (27)(900)(0.4 \times 10^{-6})(3.0)^2 = 0.087 \text{ N}$$